DATA ENGINEERING & BI ANALYSIS

HEALTHCARE EHR PROJECT

Diagnostic Insights into AI-Generated Clinical Datasets

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Methodology & Workflow

From raw Kaggle data to high-performance Power BI reports.

DATA PIPELINE & ENGINEERING

Cleaning & Excel

Establishing unique patient identifiers, deduplication, and demographic validation of gender, age, and clinical titles for structural integrity.

DAX Optimization

Architecting custom measures for Year-over-Year billing growth and condition-specific averages to drive granular performance insights.

Power BI Reporting

Developing multi-page dashboards featuring interactive filters and segmented revenue analysis by temporal and patient demographics.

DEMOGRAPHIC DISTRIBUTION

31%

Elderly Utilization (65+)

Key Demographic Markers

- Gender: Near-perfect parity (50.05% M / 49.95% F)
- Most Frequent Age: 38 Years
- Pediatrics: Minimally represented (0.21%)
- Age Range: 13 to 85 Years

CLINICAL DATASET LANDSCAPES

Medical Conditions

Uniformly distributed across the dataset (16–17% each). No single condition dominates, enabling balanced comparative analysis of disease prevalence.

Clinical Test Results

Data is split evenly into thirds: **Normal (33.06%)**, **Abnormal (33.56%)**, and **Inconclusive (33.38%)**, suggesting controlled generation.

AVERAGE BILLING BY CONDITION

Medical Condition	Average Billing Amount ($)	Variance Analysis
Obesity	25,860	Highest Average Billing
Diabetes	25,695	Consistent with mean
Asthma	25,687	Low variation
Hypertension	25,556	Predictable trend
Arthritis	25,529	Minimal cost diff
Cancer	25,215	Unexpectedly Lowest

⚠ **Finding: Highly consistent billing (~25.2K–25.9K) indicates standardized AI simulation rather than real-world variability.**

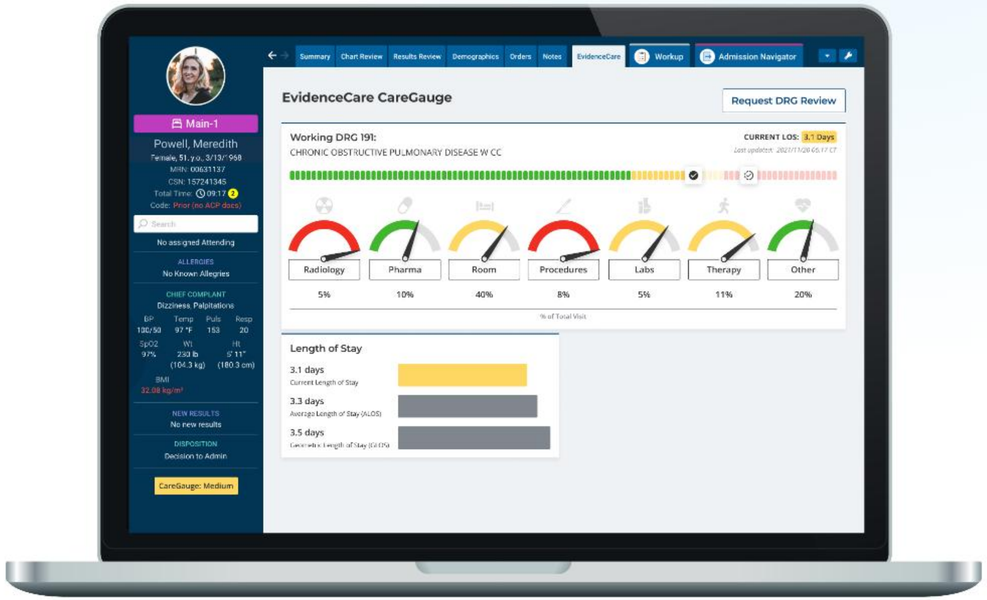
HOSPITALIZATION: LENGTH OF STAY (LOS)

Standardized Inpatient Trends

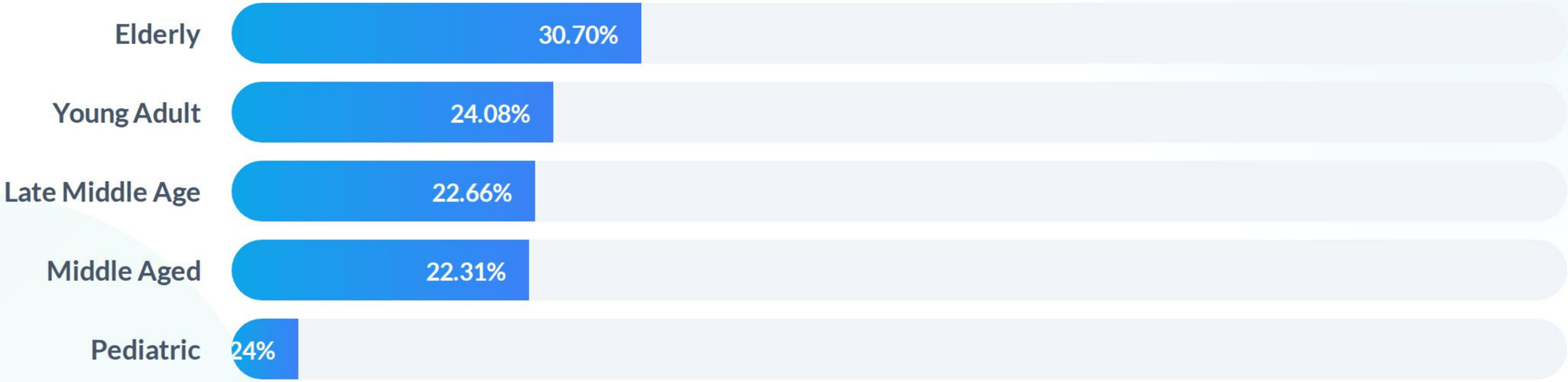
The length of stay is nearly uniform across disparate conditions, a significant **red flag** in clinical data analysis:

- **Asthma, Cancer, Diabetic:** Constant at 16 Days
- **Obesity, Hypertension, Arthritis:** Constant at 15 Days

This lack of variability confirms the AI-generated origin and limits the representation of real clinical complexity.



ABNORMAL TEST RATES BY AGE



Correlation: Abnormal test results show a positive correlation with age, identifying the elderly as the highest clinical risk group.

"The uniform distribution of clinical results and financial metrics suggests a standardized simulation model. Real-world medical data typically exhibits significantly higher variance in Cancer billing and emergency hospitalization."

— Data Integrity Analysis Summary

Project Summary

Primary Tools

Excel, Power BI

Data Source

Kaggle (AI-Generated)

Analysis Grounded in Rigorous Data Cleaning & Structured DAX Engineering

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IMAGE SOURCES

https://evidence.care/wp-content/uploads/2022/05/CareGauge-Laptop-Mockup.png

Source: evidence.care